Etsy, 2024: nearest neighbours join keyword search

This weekend: Friday 08:00, the video store

Each search and each product is a list of numbers; the products nearest by inner product join the shortlist.

Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024; preprint arXiv, June 2023.

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On popular searches a plain version found 32% of the products bought; keyword search 63%.

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What Etsy decided

A compressed index: under 4% of the nearest products missed, lookups within 20 ms.

Jha, Subramaniyam, Benjamin, Taula (Etsy), IEEE FMLDS 2024; preprint arXiv, June 2023.

Spotify, 2024: podcasts reveal audiobook tastes

This weekend: Friday 10:00, the matchmaker

Each listener and each audiobook is 128 numbers, learned from what people streamed, podcasts included.

De Nadai and others (Spotify), Companion Proceedings of the Web Conference (WWW) 2024.

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What Spotify decided

In production for every eligible user; the narrower catalogue to be revisited.

De Nadai and others (Spotify), Companion Proceedings of the Web Conference (WWW) 2024.

Adyen, 2025: a better bandit approves more payments

This weekend: Friday 12:00, the bandits

Most payments can reach the bank in 2 to 15 ways. Epsilon-greedy picks the best bet, and now and then another.

Vangara and Egg (Adyen), OARS workshop, KDD 2025. Egg (Adyen), Off-policy Evaluation for Payments at Adyen, arXiv, January 2025.

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What Adyen decided

Rules first: contracts and card network rules remove options before the bandit picks.

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LinkedIn, 2024: bigger steps instead of longer training

This weekend: Friday 14:00, gradient descent

Models rank the feed, jobs and ads. The feed's has billions of parameters, trained on several GPUs.

Borisyuk and others (LinkedIn), LiRank: Industrial Large Scale Ranking Models at LinkedIn, KDD 2024.

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What LinkedIn decided

Longer training would slow experiments: triple the learning rate instead.

Borisyuk and others (LinkedIn), LiRank: Industrial Large Scale Ranking Models at LinkedIn, KDD 2024.

Which technique answers which question

The business question	*The technique*	*The cases*
Which products, or people, are nearest to this one?	Nearest neighbours	***Etsy, 08:00*** *also Airbnb, Snap*
Which unseen product will this client want?	Matrix factorization	***Spotify, 10:00*** *also Pinterest, Uber*
Which option to use, while still learning which works?	Bandits	***Adyen, 12:00*** *also Amazon, Dream11*
How to learn the numbers from billions of examples?	Gradient descent	***LinkedIn, 14:00*** *also Kuaishou*

Airbnb, arXiv 2026. Snap, SIGIR 2023. Pinterest, KDD 2022. Uber, engineering blog 2023. Amazon, AWS blog 2026. Dream11, arXiv 2023. Kuaishou, arXiv 2025.